Yahye Bashir

Robotics Software Engineer — AI Researcher

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Summary

Early-career Robotics Engineer with a research-oriented focus on robotic manipulation and learning-based autonomy. Currently working on manipulation systems that combine robot learning, motion planning, and simulation-driven validation using ROS 2, MoveIt 2, and NVIDIA Isaac Sim.

My primary interests lie in robot learning for manipulation, vision-language models for embodied intelligence, and building end-to-end systems that enable robots to perform real-world tasks such as pick-and-place, household assistance, and object interaction. I am particularly motivated by emerging directions in Physical Intelligence (π0) and platforms such as NVIDIA Isaac GR00T, and I aim to contribute to advancing manipulation capabilities within the Turkish robotics ecosystem.

Professional Experience

Robotics Software Engineer — MORIA Project Bursa Technical University

Robotics Technologies and Intelligent Systems Research Center (ROTASAM) 04/2025 – Present

- Developed and maintained simulation models (digital twins) of manipulation platforms using Gazebo and NVIDIA Isaac Sim to support task development and testing.
- Integrated dual-arm manipulators and multi-finger robotic hands with ROS 2, MoveIt 2, and ros2_control, enabling reliable execution of pick-and-place and manipulation tasks.
- Designed motion planning pipelines for single- and dual-arm manipulation, focusing on collision avoidance, coordination, and repeatable task execution.
- Built a software-in-the-loop (SIL) workflow to test manipulation tasks, control interfaces, and motion plans before deployment on physical hardware.
- Worked closely with researchers and engineers to implement task-level manipulation behaviors and translate experimental ideas into stable ROS 2-based systems.

Education

Bursa Technical University 2025 – 2027

M.Sc. Mechatronics Engineering (Robotics) — GPA: 3.13

Northwestern University (Coursera)

Modern Robotics Specialization — Kinematics, Dynamics, Control, Planning, Estimation

Karabuk University 2019 – 2024

B.Sc. Mechanical Engineering

Selected Projects

- **Reinforcement Learning for Quadruped Locomotion** — Trained PPO-based locomotion policies in NVIDIA Isaac Sim / Isaac Lab for a custom quadruped robot, focusing on stable gait generation, velocity tracking, and robustness to terrain disturbances.
- **RL-Based Robotic Manipulation (UR10e + Robotiq)** — Formulated reach-and-lift manipulation tasks as Markov Decision Processes and trained reinforcement learning policies in ROS 2-based simulation environments.

- **Dual-Arm Manipulation in Simulation** — Integrated a dual-arm robotic system in NVIDIA Isaac Sim; developed coordinated pick-and-place behaviors with collision-aware motion planning and grasp execution.
- **Vision-Guided Manipulation (Franka Panda)** — Built a ROS 2 pipeline combining OpenCV-based perception with MoveIt 2 motion planning for color-based object sorting.
- **Autonomous Mobile Robot Navigation** — Implemented mapping, localization, and obstacle avoidance using the ROS 2 Navigation stack in Gazebo.

Internships

Control Engineer Intern — Karabük Teknokent 08/2022 – 09/2022

- Modeled helicopter flight dynamics using MATLAB/Simulink and evaluated closed-loop control behavior in simulation.
- Designed and tuned automatic control systems, improving simulated stability and response time by approximately 20%.

Mechanical Engineering Intern — Drone Market 09/2022 – 10/2022
Assisted with drone maintenance, calibration, and basic flight control testing, gaining hands-on exposure to UAV hardware systems.

Mechanical Engineering Intern — Serhat Haddecilik 09/2023 – 10/2023
Short-term industrial internship providing exposure to large-scale manufacturing, quality control, and production workflows.

Technical Skills

Robotics & Autonomy: ROS 2 (rclcpp/rclpy), ros2_control, MoveIt 2, Navigation2, Gazebo, NVIDIA Isaac Sim, Isaac Lab
Programming & Software: C++, Python, CUDA, MATLAB, CMake, Colcon
AI / Machine Learning: PyTorch, TensorFlow, OpenCV, Reinforcement Learning , imitation learning
Simulation & Modeling: Digital Twins, Physics-based Simulation
Systems & Tools: Linux, Git, Embedded Systems
Mechanical & Design: SolidWorks, Fusion360

Volunteer Experience

Robotics Software Engineer (Volunteer) — Bursa Space, Aviation & Defense Systems 09/2024 – 03/2025

- Developed an autonomous UAV platform for the Teknofest Fighting UAV Competition.
- Implemented a ROS 2-based autonomy stack integrating PX4, PID control, computer vision, and AI-based target detection.
- Designed mission planning, path planning, and autonomous execution pipelines using ArduPilot.
- Collaborated within a multidisciplinary engineering team across perception, control, and flight systems.

Certifications

- ETH Zurich — Autonomous Mobile Robots (AMR)
- DeepLearning.AI & Stanford — Machine Learning Specialization